

Dr. Zhe Wang

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EDUCATION

Ph.D. Statistics 2020

University of Connecticut, Storrs, CT

- Ph.D. research in sequential analysis under the direction of Prof. [Nitis Mukhopadhyay](#).
- Dissertation title: [Sequential Estimation Methodologies with Observations Gathered in Groups: Theory, Practice, and Data Analysis](#).

B.S. Statistics 2016

Beijing Normal University, Beijing, China

ACADEMIC APPOINTMENT

Denison University

Assistant Professor of Data Analytics

Aug. 2020–Present

PUBLICATIONS

Peer-Reviewed Articles

- Beaton, M. E., & **Wang, Z.** (2025). Generational change in the conditioning of coda /r/ manner of articulation in Lorain Puerto Rican Spanish. *Studies in Hispanic and Lusophone Linguistics*, 18(2), 261–290.
- Hu, J., Alanazi, I., & **Wang, Z.** (2024). Estimating a function of the scale parameter in a gamma distribution with bounded variance. *Sequential Analysis*, 43(3), 301–316.
- Mukhopadhyay, N., Hu, J., & **Wang, Z.** (2024). Second-order asymptotics for comparing treatment means from purely sequential estimation strategies under possible outlying observations. *Communications in Statistics—Simulation and Computation*, 53(3), 1330–1355.
- Mukhopadhyay, N., & **Wang, Z.** (2020). Purely sequential FWCI and MRPE problems for the mean of a normal population by sampling in groups with illustrations using breast cancer data. *Sequential Analysis*, 39(2), 176–213.
- Mukhopadhyay, N., & **Wang, Z.** (2020). Purely sequential estimation problems for the mean of a normal population by sampling in groups under permutations within each group and illustrations. *Sequential Analysis*, 39(4), 484–519.
- Mukhopadhyay, N., & **Wang, Z.** (2019). A general theory of purely sequential minimum risk point estimation (MRPE) of a function of the mean in a normal distribution. *Sequential Analysis*, 38(4), 480–502.

Manuscripts Under Review

- Almasri, B., Abed, M. A., Almasri, T., Khader, Y. S., Almasri, L., & **Wang, Z.** (2026). Development and validation of a continuous metabolic syndrome severity score in the Jordanian population: A principal component analysis approach. Manuscript submitted to *Journal of Advanced Nursing*.
- **Wang, Z.**, Waterbury, A., & Pham, B. (2026). Exact and Bayesian inference for the overlap coefficients of generalized gamma distributions. Manuscript submitted to *Communications in Statistics—Simulation and Computation*.
- **Wang, Z.**, Zhuang, Y., & Pham, B. (2026). Confidence interval estimation for the Matusita overlap coefficient under the generalized gamma distribution. Manuscript submitted to *Sequential Analysis*.

Manuscripts in Preparation

- Agarwal, P., **Wang, Z.**, Dharmarajan, K., & Mazumdar, M. A tutorial and application of latent class and latent transition analysis on electronic medical records and Medicare outcomes data. Manuscript in preparation.

CONFERENCE PRESENTATIONS

1. *Second-Order Asymptotic Property of the Sequential Estimation of Overlap Coefficient*. Invited Speaker, **International Day of Women in Statistics and Data Science**, Virtual. Oct. 2026
2. *Quantitative Findings of AI Sentiment in the Classroom*. Invited Speaker, **AALAC Workshop on GenAI and Data Analysis**, Middletown, CT. July 2026
3. *Sequential Estimation and Boundary Correction for the Overlap Coefficient*. Invited Speaker, **International Workshop on Sequential Methodology**, Washington, DC. June 2026
4. *Sequential Analysis in the Undergraduate Classroom*. Panel session organizer and speaker, **International Workshop on Sequential Methodology**, Washington, DC. June 2026
5. *Innovations and Challenges in Statistics Education at Liberal Arts Colleges*. Panel session organizer and speaker, **Women in Statistics and Data Science**, Cincinnati, OH. Nov. 2025
6. *Practical Visualization and Interpretation of Generalized Linear Mixed-Effects Models in Linguistic Research*. Speaker, **Denison Scientific Association**, Granville, OH. Oct. 2025
7. *Diverse Drivers of Change in a Language Contact Context: Puerto Rican Rhotics in Lorain, Ohio*. Co-author, **29th Conference of Spanish in the US & 14th Conference of Spanish in Contact with Other Languages (SIUS25)**, San Antonio, TX. April 2025
8. *Rhotic Change and Maintenance in US Puerto Rican Spanish*. Co-author, **Hispanic Linguistics Symposium (HLS)**, Provo, UT. Oct. 2023
9. *Inference on the overlapping coefficients*. Invited Speaker, **The 36th New England Statistics Symposium**, Boston, MA. June 2023

10. *Latent Class and Latent Transition Analysis*. Session organizer and speaker, **Women in Statistics and Data Science**, *St. Louis, MO*. Oct. 2022
11. *A Review of Missing Data Approaches and Practices in Large Nationally Representative Survey Databases*. Co-author, **Joint Statistical Meetings**, *Washington, DC*. Aug. 2022
12. *You Will Never Get Bored as a Statistician*. Speaker, **Denison University Data Analytics Research Talk**, *Granville, OH*. April 2022
13. *Generational Differences in Coda Liquid Production in Lorain*. Co-author, **10th International Workshop on Spanish Sociolinguistics**, *Atlanta, GA*. April 2022
14. *Ramp Detection Methods for Renewable Energy Integration*. Invited Speaker, **The 34th New England Statistics Symposium**, *Providence, RI*. Sept. 2021
15. *Change Point Detection Methodologies and Applications*. Session organizer and chair, **The 38th Annual Quality and Productivity Research Conference**, *Virtual*. July 2021
16. *Minimum Risk Point Estimation for the Mean of a Normal Population by Sampling in Groups*. Invited Speaker, **Upstate New York Statistics Conference**, *Virtual*. March 2021
17. *How to Find the Minimum Risk Point Estimation of Treatment under the Normality Assumption*. Invited Speaker, **The 7th International Workshop in Sequential Methodologies**, *Binghamton, NY*. June 2019
18. *Sequential Fixed-Width Confidence Interval Estimation for the Mean of a Normal Population: Sampling in Groups*. Invited Speaker, **The 36th Annual Quality and Productivity Research Conference**, *Washington, DC*. June 2019

INVITED LECTURES

1. *Sequential Design for Microarray Experiments*. Invited Speaker, **STAT 8750, Department of Statistics, The Ohio State University**, *Columbus, OH*. March 2023
2. *A Gentle Introduction to Sequential Adaptive Design*. Invited Speaker, **Biostatistics Seminar, Icahn School of Medicine at Mount Sinai**, *New York, NY*. May 2022
3. *Baseball Analytics Workshop*. Invited Short Course Instructor, **UConn Sports Analytics Symposium**, *Virtual*. Oct. 2020

GRANTS, FELLOWSHIPS, AND AWARDS

- Mu Sigma Rho Early Career Undergraduate Impact Award 2026
- Anderson Science Research Program (Faculty Stipend), Denison University 2022
- Denison University Research Foundation, Denison University 2022–2023, 2026–2027

UNIVERSITY SERVICE

- **Denison Scientific Association**, Organizer 2025–2026
Organize monthly interdisciplinary scientific research talks for faculty and students.
- **ASA DataFest at Denison**, Faculty Lead 2026
Recruited judges and volunteers from JPMorgan Chase and mentored 13 student teams during the competition.
- **Data Analytics Faculty Search Committees**
 - Visiting Assistant Professor searches 2021, 2026
 - Tenure-track Assistant Professor searches 2022, 2025
- **Curriculum Development Committees**
 - Applied Mathematics Major 2021
 - Data Analytics Minor 2025–2026
 - B.S. in Data Analytics 2025–2026

PROFESSIONAL LEADERSHIP AND SERVICE

- **American Statistical Association, Columbus Chapter**, *Columbus, OH* 2026–2028
Vice President (2026); President (2027); Past President (2028).
- **Math Alliance**, Denison University Representative, *Granville, OH* 2021–Present
Connect students from underrepresented or underserved backgrounds with mentoring and preparation for doctoral study in the mathematical sciences.
- **Connecticut Sports Analytics Symposium**, *Storrs, CT* 2026
Conference Committee; Data Challenge Judge.
- **American Statistical Association Mentoring**
 - Mentor, Columbus Chapter Mentoring Program 2025–2026
 - Mentor, Section on Statistics and Data Science Education Mentoring Program 2022–2023
- **Florence Nightingale Day: Statistics and Data Science**, *Columbus, OH* 2021–2023
Conference Committee; Director of Analytics.
- **ASA Undergraduate Statistics Research Project Competition**, Judge 2020

SELECTED STUDENT MENTORING

- Bao Pham, *undergraduate research student at Denison University* 2025–Present
 - Research on inference for generalized gamma distributions; progressed from checking derivations to developing new results.
 - Co-author on Wang, Waterbury, & Pham (2026), currently under review.

- Wanxin Zhou, *undergraduate student at The Ohio State University* 2025–Present
 - Formal mentee through the American Statistical Association (ASA) Columbus Chapter Mentoring Program.
- Minh Le, Andrew Nguyen, and Dong Dong, *undergraduate students in DA352* Fall 2025
 - Project title: [A Decade of Decline: Statistical Modeling of Bank of America’s Physical Branch Closures](#).
 - Won Third Prize in the American Statistical Association [Undergraduate Class Project Competition](#).
- Vimal Rao, *Teaching Assistant Professor at the University of Illinois Urbana-Champaign* 2022–2023
 - Formal mentee through the American Statistical Association (ASA) Section on Statistics and Data Science Education Mentoring Program.
 - Met monthly on academic job search and early-career development.
- Mark Raney, *Anderson Science Summer Research Scholar* Summer 2022
 - Project title: [Assessing the similarity of countries’ COVID-19 experiences via time-series clustering](#).
- Riley Coburn, *Anderson Science Summer Research Scholar* Summer 2022
 - Project title: [Change Point Detection with Dynamic Linear Modeling on COVID-19 Time Series Data](#).
 - Poster presented at the 2023 International Conference on Health Policy Statistics (ICHPS).
 - Now a Ph.D. candidate in Statistics at The Ohio State University.
- Yuhan Fu, *DURF Summer Research Student* Summer 2022
 - Project title: [Bayesian Online Changepoint Detection on COVID-19 Data](#).
 - Poster presented at the Nebraska Conference for Undergraduate Women in Mathematics.
 - Now a Ph.D. candidate in Mathematics at the University of Nebraska–Lincoln.
- Aditi Singh, *senior research student* Spring 2022
 - Project title: [Computational Analysis of the Transformation of Former Colonial Countries](#).
 - Won Third Prize in the American Statistical Association [Undergraduate Statistics Research Project Competition](#).

PROFESSIONAL MEMBERSHIPS

- American Statistical Association (ASA)
- Institute of Mathematical Statistics (IMS)
- New England Statistical Society (NESS)